

NLP Overview: Models, Advantages & Disadvantages

A structured overview of topics, models, and their trade-offs.

Topic / Section	Models / Approaches	Advantages	Disadvantages
1. Text Preprocessing (L01 & L02)	NLTK vs. spaCy	NLTK: Huge selection of classical algorithms; ideal for research purposes. spaCy: Extremely fast (Cython); object-oriented; optimized for production environments.	NLTK: Slower execution; steep learning curve. spaCy: Less flexible for specific classical algorithms.
	Stemming vs. Lemmatization	Stemming: Very fast and simple to implement. Lemmatization: Considers context and Part-of-Speech (PoS); provides valid dictionary base forms.	Stemming: Less accurate; often generates non-real words (over-stemming). Lemmatization: More computationally intensive and slower than stemming.
2. Language Modeling (L03)	N-Grams (Unigram, Bigram, Trigram)	Simple to understand; efficient for word prediction within clear, local contexts.	Zero-Probability Problem: Unknown sequences lead to probability zero. Data Sparsity: Requires smoothing techniques to handle unseen data.
3. Text Classification (L04 & L05)	Naive Bayes (Generative)	Computationally efficient; scales well for large datasets; serves as a strong baseline classifier	Independence assumption: Ignores word order ("naive" assumption). Faces performance issues with imbalanced data..
	Logistic Regression (Discriminative)	Models probabilities directly; learns decision boundaries between classes more precisely.	Assumes a linear relationship between features and the log-odds of the outcome.
4. Sentiment Analysis (L06)	Lexicon-based (e.g., VADER, AFINN)	No training data required; transparent/interpretable logic; effective for simple sentiment tasks.	Performs poorly with context-heavy language, sarcasm, slang, and irony.
	Machine Learning (e.g., SVM, Neural Networks)	High adaptability; capable of learning complex linguistic nuances and semantic context.	Requires large volumes of manually labeled training data for effective performance.
5. Topic Modelling (L07)	LDA (Latent Dirichlet Allocation)	Established classical standard; fast; provides high interpretability of word distributions.	Based on Bag-of-Words (ignores semantics); number of topics (K) must be predefined.
	Top2Vec & BERTopic	Captures deep semantics; can often determine the number of topics automatically; strong performance on short texts.	Top2Vec: Requires tuning of many hyperparameters BERTopic: Highly computationally intensive due to Transformer-based architecture.
6. Vector Semantics (L08)	Word2Vec (Skip-Gram / CBOW)	Captures semantic similarities (vector analogies); dense vectors generalize better than sparse ones.	Static Embeddings: A word is assigned the same vector regardless of the specific context it appears in.
7. Neural Language Models (L09)	Feed-Forward Neural Networks (FFNN)	Capable of learning non-linear relationships (resolves XOR problem); utilizes representation learning.	High computational requirements; "black-box" nature makes weight interpretation and explainability difficult.